

ABC SENIOR SECONDARY SCHOOL
CBSE CLASS XII — BOARD EXAMINATION (THEORY)

Subject: Science

Time Allowed: 3 Hours | Maximum Marks: 80 | Date: 25 May 2026

Name: _____ Roll No.: _____
Class/Section: _____

GENERAL INSTRUCTIONS

- 1. All questions are compulsory.*
- 2. Read each question carefully before answering.*
- 3. Figures to the right indicate full marks.*
- 4. Use of calculator is not permitted unless stated.*

SECTION A — MULTIPLE CHOICE QUESTIONS

Each question carries 1 mark. Select the correct option.

Q1. (1 marks)

Which of the following statements about image formation is true?

- (a) A convex mirror always forms a real, inverted and enlarged image.
- (b) A concave lens can form both real and virtual images depending on the object's position.
- (c) A concave mirror cannot form an erect or magnified image.
- (d) A plane mirror never forms an erect image.

Q2. (1 marks)

A concave mirror can form both real and virtual images depending on the object's distance from it.

- (a) T
- (b) F

Q3. (1 marks)

A concave mirror can form both real and virtual images depending on the object's distance from the mirror.

- (a) T
- (b) F

Q4. (1 marks)

Which of the following statements about images formed by lenses and mirrors is true?

- (a) A concave lens can only form a virtual image.
- (b) A convex mirror always forms an erect, enlarged real image.
- (c) An object placed between the focal point and a plane mirror will produce no image at all.
- (d) When parallel rays of light strike one side of a concave lens, they diverge as if coming from behind the lens.

Q5. (1 marks)

What is the primary purpose of a circuit breaker in an electrical system?

- (a) To convert AC to DC current.
- (b) To regulate voltage levels within circuits.
- (c) To protect against excessive currents that can cause damage or fire by interrupting the circuit flow when a fault is detected.
- (d) To store electrical energy for later use.

Q6. (1 marks)

In the context of image formation by convex and concave mirrors as well as lenses in a physics course on optics for Class XI students during Circuit Breaker (CB) period, which statement is true?

- (a) T: Convex mirror always form an erect image.
- (b) F: A concave lens can sometimes produce a real and inverted image.

Q7. (1 marks)

Which statement about concave and convex mirror image properties is true?

- (a) A Convex Mirror Always Produces an Enlarged, Erect Image.
- (b) A Concave Lens Can Form a Virtual Image Under Certain Conditions.
- (c) Only A Plane Mirror Reflects Light and Does Not Distort Images.
- (d) Both Statements Are True.

Q8. (1 marks)

Which of the following statements about lenses is correct?

- (a) A concave lens always forms a real image.
- (b) A convex lens can form both real and virtual images depending on object distance.
- (c) A convex lens only forms virtual, erect images when the object is within one focal length from it.
- (d) Both B and C are correct.

Q9. (1 marks)

Explain the key concept from the uploaded syllabus.

Q10. (1 marks)

Which of the following statements about convex lenses is true?

- (a) A Convex lens always forms a virtual, erect image.
- (b) B A real and enlarged image can be formed when an object is placed between the focal point and the lens.
- (c) C An upright image cannot be obtained with a convex lens under any circumstance.
- (d) D When parallel rays of light pass through a convex lens, they converge at one or more points called principal foci.

Q11. (1 marks)

Which of the following statements about image formation is true?

- (a) A real and erect image can be formed by a convex mirror.
- (b) Virtual images are always enlarged in size when produced by concave lenses.
- (c) Convex mirrors only form virtual, upright, and reduced images of objects placed close to them.
- (d) When an object is real and located beyond the focal point of a converging mirror or lens, it can produce both erect and inverted images.

Q12. (1 marks)

A convex mirror always forms an image that is

- (a) Smaller and erect.
- (b) Larger and inverted.
- (c) Same size as the object, real or virtual.
- (d) Small and magnified.

Q13. (1 marks)

Which of the following statements is true regarding image formation by mirrors and lenses?

- (a) A. A convex mirror always forms a virtual, erect, and reduced image.
- (b) B. All concave mirrors form real images regardless of the object's position relative to the focal point.
- (c) C. Both convex lenses and concave mirrors can produce magnified real images under certain conditions.
- (d) D. A plane mirror cannot produce a virtual or diminished image.

Q14. (1 marks)

Which of the following statements is true regarding image formation by a mirror or lens?

- (a) A. A convex mirror always forms an enlarged and erect virtual image.
- (b) B. An object placed between F and D focus in a concave mirror produces no real, inverted images on the screen.
- (c) C. Both (a) and (b) are true.
- (d) D. Only when objects are very close to convex lenses can enlarged and erect virtual images be formed.

Q15. (1 marks)

A convex lens can produce...

- (a) an enlarged and erect image if the object is placed between the focal point and the optical center of the lens.
- (b) a reduced and inverted image regardless of the object's position relative to the focal length.
- (c) no real or virtual images, only colorful patterns are formed due to dispersion.
- (d) an enlarged and erect image if the object is placed at any distance from the lens.

Q16. (1 marks)

Which of the following statements about lenses and mirrors is true?

- (a) A concave lens always forms a virtual, erect image.
- (b) We can obtain an enlarged and erect real image by using a convex mirror.
- (c) A concave mirror cannot produce a real image when the object is placed between F and C of the focal point.
- (d) An upright virtual image formed by both plane mirrors.

Q17. (1 marks)

Which of the following statements is true about a convex mirror?

- (a) A: It always forms real and inverted images.
- (b) B: It can form virtual, erect, and diminished images when used as a shaving mirror.
- (c) C: Its focal length is positive.
- (d) D: A convex lens cannot produce an enlarged image.

Q18. (1 marks)

Which of the following statements about lens and mirror optics is true?

- (a) A convex mirror always produces an erect, diminished image.
- (b) The focal length of a concave lens can be negative if it's diverging.
- (c) An object placed between the focal point and the center of a convergent mirror will produce a virtual, upright, enlarged image.
- (d) A real image formed by a convex mirror cannot be captured on a screen.

Q19. (1 marks)

Which of the following statements about concave lenses and convex mirrors is true?

- (a) A. Convex mirror always forms a virtual image.
- (b) B. A real, enlarged and inverted image can be formed by a concave mirror under certain conditions.
- (c) C. Both convex lenses and concave mirrors form only real images when the object is placed within one focal length from them.

(d) D. Concave lenses always produce an erect virtual image, while convex mirrors do not necessarily yield such an outcome.

Q20. (1 marks)

Which of the following statements about concave mirrors and convex lenses is true?

(a) A A Concave mirror always forms a virtual image.

(b) B A convex lens can form both real and virtual images depending on the object's position relative to its focal point.

(c) C We cannot obtain an enlarged, erect image with either concave mirrors or convex lenses under any circumstances.

(d) D Convex lenses always produce smaller, inverted images.

SECTION B — VERY SHORT ANSWER

Answer in 40-60 words. Each question carries 2 marks.

Q21. (2 marks)

Refer to Fig. 13.15 and compare the slopes of lines A and B on a distance-time graph to determine which vehicle is moving faster.

Q22. (2 marks)

Refer to Fig. 13.15 in your textbook where two vehicles A and B are shown on a distance-time graph.

Q23. (2 marks)

Which vehicle is moving faster based on their distance-time graph?

Q24. (2 marks)

Refer to Fig. 13.15 in the context of SCIENCE32.

Q25. (2 marks)

Refer to Fig. 13.15 in the distance-time graph exercise.

Q26. (2 marks)

Refer to Fig. 13.15 in the context of distance-time graphs for vehicles A and B.

SECTION C — SHORT ANSWER

Answer in 60-80 words. Each question carries 3 marks.

Q27. (3 marks)

Boojho initially perceived hot and cold water separately with his left and right hands, respectively. After dipping both hands into mug A (hot) for a few minutes followed by immersion in separate containers of cooler temperatures B and C before putting them together back to the same temperature-neutralized medium at mug C simultaneously after 2–3 minutes.

Q28. (3 marks)

After conducting an experiment with water in mugs A and B at different temperatures and then placing both hands simultaneously into a third mug C containing the same water, what conclusion can be drawn about temperature perception based on sensory information from each hand?

Q29. (3 marks)

Explain the principle that states when work is done on an object or by an object in a closed system without any non-conservative forces, such as friction, acting upon it. Provide examples of two changes (changes not necessarily physical) where energy conservation applies and describe why these are considered harmful to society or environment. Suggest methods for preventing each type of change.

Q30. (3 marks)

After Boojho dips his hands in mug A and B for a few minutes before putting both hands into the hot water of mug C, will he experience different sensations on each hand? Explain your answer.

Q31. (3 marks)

Consider the following scenario: Boojho performs an experiment with hot and cold water in mug C. He then dips his hands into separate cups of warm (mug A) and cooler (mug B) water for 2-3 minutes before placing both hands simultaneously back into mug C containing mixed temperatures from the initial setup.

Question: Based on Boojho's experiment, explain why when he puts his hand in each cup of warm and cool water separately followed by putting them together in a third container with hot and cold water initially present, all three hands felt as if they were immersed only in mug C. What principle does this demonstrate about the perception of temperature?

Q32. (3 marks)

After immersing your hands in mugs A and B for a few minutes each with different temperatures of water, you then simultaneously place both hands into the same third mug C containing cold water. Considering that thermal energy will transfer from warmer objects to cooler ones until they reach equilibrium:

SECTION D — CASE BASED / SOURCE BASED

Read the passage and answer the sub-questions. Each question carries 4 marks.

Q33. (4 marks)

A convex lens is a converging optical device that brings parallel rays of light to focus at one point. Conversely, a diverging or concave lens causes the incident parallel light rays to spread out and appear as if they are emanating from a virtual focal point behind the mirror.

Q34. (4 marks)

A convex lens is used in a simple magnifying glass. When an object is placed at various distances from the lens:

(a) At what distance does the image appear largest and upright?

(b) What happens to the size of the image when the object moves closer to this point than 25 cm?

Q35. (4 marks)

A convex lens always forms a real and inverted image when the object is placed at any distance from it. A concave mirror can form both real and virtual images depending on the position of the object relative to its focal point.

SECTION E — LONG ANSWER

Answer in detail with diagrams where necessary.

Q36. (5 marks)

An electromagnet is created when an electric current passes through a coil of wire wrapped around a piece of iron. This setup allows for the temporary creation and retraction of magnetic force, which can be used to pull or push objects made from ferromagnetic materials like steel.

When an electromagnet is activated by closing a circuit with a screw attached to the iron strip (also known as a solenoid), it generates magnetic force strong enough to pull on metal objects, such as pulling down a bell's clapper. This action demonstrates how electricity can be converted into mechanical motion.

The electromagnetic effect is reversible due to the nature of circuits; when the iron strip breaks contact with the screw, it opens the circuit and stops the current flow. This interruption causes a release in magnetic force allowing objects like clappers or other metal strips attached to the system to be released.

The electromagnetic effect can also produce sound when used with devices such as bells. When an iron strip is pulled towards a magnet, it strikes against another object (like a bell clapper), causing the bell to ring.

This process of using electromagnets for mechanical motion and sound production highlights their practical applications in everyday devices. It also serves as an introduction to understanding how electricity can be harnessed not just for lighting or power but for physical actions.

Q37. (5 marks)

Describe the process by which an electromagnet is created and used in this lesson. Include details on how it operates, its effect on breaking a circuit when activated, and why such behavior might be advantageous for certain applications.

SECTION F — EXTENDED RESPONSE

Answer with proper derivation / explanation. (8 marks)

Q38. (8 marks)

A concave mirror can produce both real and virtual images depending on the object's distance from the mirror. When an object is placed between the focal point (F) and the mirror, a magnified erect image is formed which appears to be behind the mirror.

The characteristics of this type of image are as follows:

- It cannot be projected onto a screen because it's virtual; only real images can be captured on screens.
- The ray diagram for an object placed between F and P (point) would show reflected rays converging to form the erect, magnified image behind the mirror.

To understand this concept better, let us consider a detailed explanation with steps/examples:

1. Identify that concave mirror is curved inward like the inside of a bowl and has focal points (F) where parallel rays converge after reflection.